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Modal Analysis of Chladni Plate Using Cymatics

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Abstract

This work aims at demonstrating nodes and antinodes at various frequencies of vibrations. Chladni plate is used for this purpose. When the plate is excited because of vibrations from a vibrator source, the sand of the plate creates specific patterns. These patterns are related to the excitation frequency. The sand on the plate moves away from antinodes where the amplitude of the standing wave is maximum and towards nodal lines where the amplitude is minimum or zero, forming patterns known as Chladni figures. The formation of patterns depends on material properties, geometry of plate, and thickness of plate and frequency/vibration pattern of the vibrator. The experimental setup consisted of a aluminum rectangular plate of 16 cm × 16 cm and aluminum circular plate of diameter 16 cm are having

thickness of 0.61 mm placed over a mechanical vibrator (GelsonLab HSPW-003), which was driven by a sine wave signal generator (Ningbo Hema scientific). The procedure consisted in sprinkling powder on the plate and sweeping the frequency of the function generator until a constant pattern was obtained. Different shapes formed by sand particle on the plate at different frequencies will be captured by the camera. Modal analysis is done using ANSYS workbench for validation purpose. Proposed work is at TRL (technology readiness level) of 3, research to prove feasibility [TRL of 3 of NASA 1 to 9 TRL levels]. As basic research is carried out to verify the use of Chladni Patterns to relate the frequency of vibration and the patterns generated. There is a possibility to work on plate geometry and material properties to relate the patterns generated with the vibrations and the chladni patterns.

Introduction

Cymatics is the branch that studies the visible effects of the sound and vibration. Derived from Greek word 'kyma', cymatics investigates wave phenomena and it is used to express the form of the responses of sound in diverse frequencies. In cymatics, Hans Jenny points out that "Typically the surface of a plate, diaphragm or membrane is vibrated, and regions of maximum and minimum displacement are made visible in a thin coating of particles, paste or liquid" [1]. In 1787, the musician and physicist Ernst Florence Friedrich Chladni from Leipzig made an interesting discovery. He noticed that when he tried to excite a metal plate with the bow of his violin, he could make sounds of different pitch, depending on where he touched the plate with the bow [2]. The general purpose of chladni experiment is visualization of acoustical stationary waves in a plate [3]. The vibrating plate which has been consecutively studied and popularized by Chladni and chladni is classical two-dimensional (2D) wave systems. The Chladni experiment has played key role in various fields of research like seismology, quantum billiards, musical instruments, and nano-mechanics [4, 5, 6]. These days, the Chladni test is performed by utilizing an electronically controlled mechanical oscillator to locally drive the plate with a variable frequency. It is beneficial to take note of that when the measures of sand particles are under 0.1 mm, the grains may relocate to the antinodes to form the so-called inverse Chladni patterns [7].

Currently, new area of research is experiment on inverse chladni patterns, Chladni nodal line patterns have been widely exhibited in popular science [8]. One method, for acoustic manipulation of a particle kept on a plate, is based on Chladni effect [9]. Chladni effect refers to the phenomenon that sand or salt particles on a vibrating plate move to and collect at the nodal lines of the plate. The location of these nodal lines changes with the change in frequency of the vibrations given to the plate [10]. Application of Chladni's method further indicates that plates with different shapes under the influence of vibration can present diverse patterns on the surface of the plate. The natural frequencies at which thin plates vibrate is vital to the visualization of nodal lines formed by the Eigen modes. The ideal method of analyzing these patterns is by solving the inhomogeneous Helmholtz equation using proper boundary conditions. However, it is difficult and time consuming to accurately employ this method, particularly in the case of vibrating plates with irregular open boundaries [11]. Amore explains a method for solving the Helmholtz equation using what he called "little sinc functions". Method is explained by Amore works for inhomogeneous irregular and/or membrane with fix boundary conditions. Only Muller and Gander used finite element methods (FEM) to find the eigen values of the homogenous Helmholtz equation; this approach excellent reproduce the Chladni figures for the rectangular plate case but the final product requires the use of a computer to make it useful [12, 13, 14].

The problem of free transverse vibrations of a rectangular plate is one of the advanced in year problems. Soon in the 19th century, owing to aesthetics, experimental work of Chladni figures [15]. The main purpose of a rectangular plate as an element of structure mechanics and engineering applications gave rise to a great number of publications devoted to studying the vibration problem using discontinuous outlook. One of them is the Ritz method [10], which was early discussed for solving the problem of vibrations of a plate with free edges [11]. Concerning alternative experimental modal test methods, Bergen and Pechersky did a experiment on submerged composite square plates using Digital Speckle Pattern Interferometry (DSPI), which is like holographic interferometry [16]. Askari performed modal analysis tests on circular plates dipped in a fluid at different depths; the plate responses were scaled with a laser Doppler vibrometer through a transparent wall at the base of the tank, and the experimental results were used to validate a theoretical model.

To guess a mode shape from the experiment work, only 3 to 5 measurements (equally distributed along the radius) were showed for the axisymmetric modes with 1 and 2 nodal circles. The Chladni technique has been proved to visualize or visual effect for the nodal lines with a high spatial resolution [17]. In this paper, shapes are formed on chladni plate at different frequencies are validated by ANSYS workbench for validation purpose.

Fundamental Equation for Rectangular Plate

Assuming a thin flat plate excited by a time-harmonic source $F(r')$, the response function $\Psi(r, r', \tilde{k})$ of the vibrating plate given by Equation (1);

$$(\nabla^2 + \tilde{k}^2)\Psi(r, r', \tilde{k}) = F(r') \quad (1)$$

Where, $\tilde{k} = k + iy$, y is the damping coefficient and k is the driving wave number

Equivalent deflections found for each mode shapes are given in Equation (2);

$$\begin{aligned} V_{max} = & \frac{D}{2} \iint_A \left\{ \left(\frac{\partial^2 W}{\partial x^2} + \frac{\partial^2 W}{\partial y^2} \right)^2 \right. \\ & \left. - 2(1-\nu) \left[\frac{\partial^2 W}{\partial x^2} \frac{\partial^2 W}{\partial y^2} - \left(\frac{\partial^2 W}{\partial x \partial y} \right)^2 \right] \right\} dx dy \quad (2) \\ = & \frac{1}{2} K_{eff} [W_0(x_0, y_0)]^2 \end{aligned}$$

Where, $W_0(x_0, y_0)$ is the point on the square plate which has the maximum deflection and (x_0, y_0) is the location of that point. The maximum deflection point differs for different mode shapes.

The effective coupling efficiency of the normalized power transferred from the point source to the vibrating plate or object under a time-harmonic condition, given by Equation (3);

$$\eta(r, \tilde{k}) = \left| \sum_n k_n^2 \cdot a_n(r, \tilde{k}) \cdot \phi_n(r') \right|^2 \quad (3)$$

Where k is the driving wave number, k_n is the eigen values close to the driving wave number, $\phi_n(r')$ is the eigen modes. Applying the Greens function, the response function excited by any general source $F(r')$ [14] is given by;

$$\Psi(r, r', \tilde{k}) = \int_V G(r, r', \tilde{k}) F(r') d^3 r' \quad (4)$$

If $y \ll k$, the Green's function can further be expressed with the Eigen modes [14] given as;

$$G(r, r', \tilde{k}) = \sum_n \frac{\phi_n^*(r') \cdot \phi_n(r)}{(k^2 - k_n^2) + 2iyk'} \quad (5)$$

Where, $\phi_n(r)$ are eigen modes and k_n are eigen values. Substituting Equation (5) into Equation (4), the expression for normalized response function $\Psi(r, r', \tilde{k})$ is given by Equation (6);

$$\Psi(r, r', \tilde{k}) = \sum_n a_n(r, \tilde{k}) \phi_n(r) \quad (6)$$

Considering a piece of square plate, the eigen modes are given by Equation (7);

$$\phi_{n_1, n_2}(x, y) = \frac{2}{a} \cos\left(\frac{n_1 \pi}{a} x\right) \cos\left(\frac{n_2}{a} y\right) \quad (7)$$

Where $n_1 = 0, 1, 2, 3, \dots$ and $n_2 = 0, 1, 2, 3, \dots$, are the eigen values the eigen modes $\phi_{n_1, 2}(x, y)$ are given Equation (8);

$$k_{n_1, n_2} = \frac{\pi}{a} \sqrt{n_1^2 + n_2^2} \quad (8)$$

With a point source at the middle, the driving function can be written in Equation (9);

$$F(r') = F_0 \delta(x' - a/2) \delta(y' - a/2) \quad (9)$$

Where, F_0 is the amplitude of the driving source Equation (10) is the forth-order wave equation describing flexural or bending waves in the thin plate vibrating under the influence of electro-mechanical oscillator [15].

$$\rho t \frac{\partial^2 u}{\partial t^2} + B_{xx} \frac{\partial^4 u}{\partial x^4} + 2B_{xy} \frac{\partial^2 u}{\partial x^2 \partial y^2} + B_{yy} \frac{\partial^4 u}{\partial y^4} = 0 \quad (10)$$

Where,

$$B_{xx} = \frac{E_{xx} t^3}{12\rho(1-\nu_{xx}^2)} \text{ and } B_{yy} = \frac{E_{yy} t^3}{12\rho(1-\nu_{yy}^2)} \quad (11)$$

Where, t is the flat plate thickness, E_{xx} , ν_{yy} , B_{xx} and B_{yy} are the elastic constant along the symmetrical regions in the x and y axis of the square/rectangular plates. For thin rectangular/square plates under the influence of vibration, Kaczmarek et al. [16] proposed a relationship for the frequency of modes (Hz) expressed in Equation (12);

$$f = \sqrt{\left(\frac{n}{l}\right)^2 + \left(\frac{m}{w}\right)^2} \quad (12)$$

Where, l is length of the plate, w is width of the plate

Fundamental Equation for Circular Plate

Postulates to a thin circular plate of radius R and excitation at the center. Let φ represents the amplitude at any inside point the plate, then the mathematical expression in polar coordinates (r, θ) of the proposed model for this case-

$$\psi(r, \theta) = \frac{A}{r} e^{ikr} + \frac{A}{r} e^{-ik(r+\varphi)} + A \int_0^{2\pi} \frac{e^{ik|\vec{r}-R\hat{\theta}|}}{|\vec{r}-R\hat{\theta}|} \cos(kR\theta) d\theta \quad (13)$$

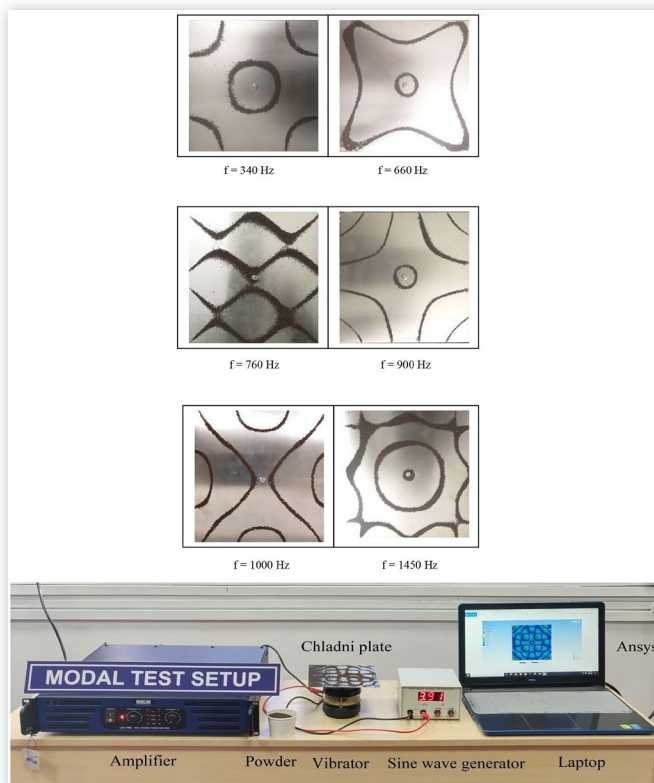
Where, A - amplitude of the wave excitation point and k is the wave number permitted by above postulate, is the phase change mentioned in postulate $2\pi r = n\lambda$. The first and second term shows the outgoing, and reflected wave. Look on that the reflected wave have a phase shift and the third term represents the infinite sum of waves around the plate modulated by an stationary wave characterize by the cosine function.

Methodology

Mode Shape for Rectangular Plate

The experimental setup consisted of a aluminum rectangular plate of $16 \text{ cm} \times 16 \text{ cm}$, 0.61 mm thickness and drilled hole of

FIGURE 1 Experimental set up for modal analysis of chladni plate

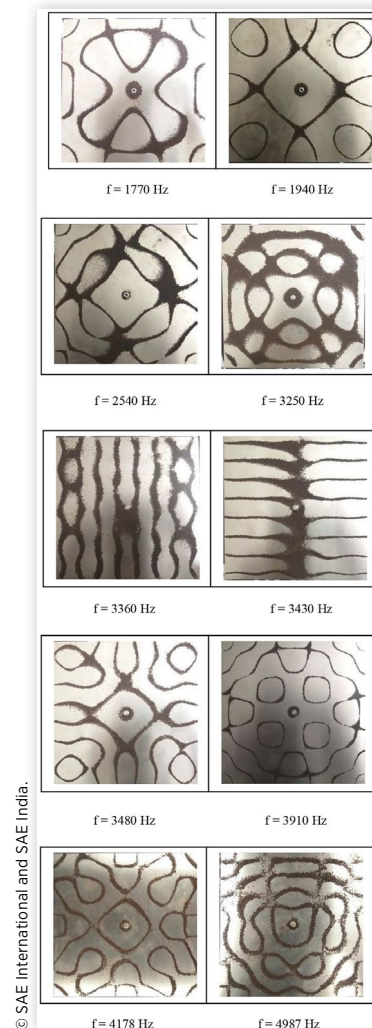


3 mm at centre of plate & a aluminum circular plate of 16 cm diameter, 0.61 mm thickness and drilled hole of 3 mm at centre are kept over a mechanical vibrator (GelsonLab HSPW-003), which was driven by a sine wave signal generator (Ningbo Hema scientific) and amplifier. Tea powder is used for making patterns. The procedure consisted in sprinkling powder on the plate and changing the frequency of the sine wave signal generator until a constant pattern was obtained. Different shapes formed by tea powder particle on the plate at different frequencies will be captured by the camera. Modal analysis is done using ANSYS workbench for validation purpose.

Experimentation

Experiment performed on the rectangular chladni plate and circular chladni plate. Obtained various mode shapes on chaldni plate at the various frequency. When natural frequency of plate matches with excitation frequency then resonance will occur and mode shape will be formed on the chaldni plate. Varied frequency form 0 Hz to 5000 Hz by sine wave signal generator and obtained 16 mode shape at rectangular plate and 3 mode shape at circular plate.

FIGURE 2 Experimnetal mode shapes for rectangular plate



Mode Shape for Circular Plate

FIGURE 3 Experimental mode shapes for circular plate

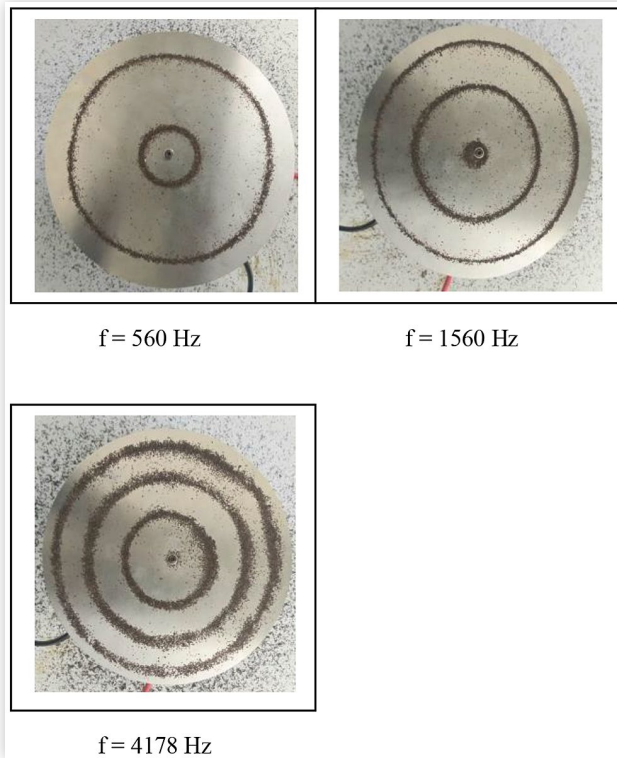


FIGURE 4 Meshed rectangular plate

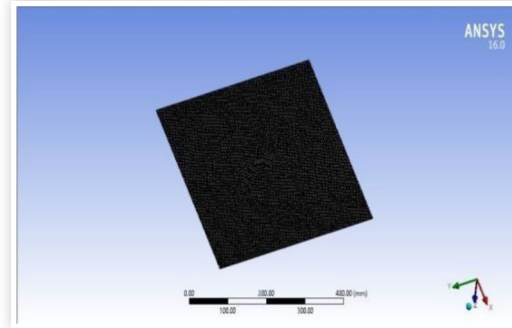
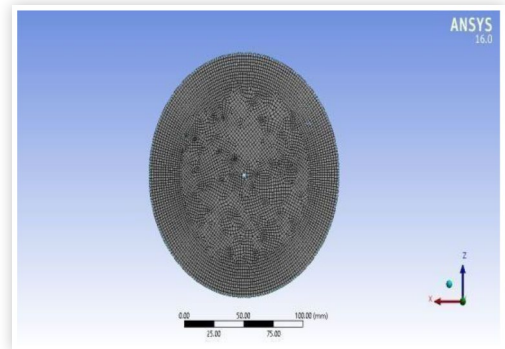


FIGURE 5 Meshed circular plate

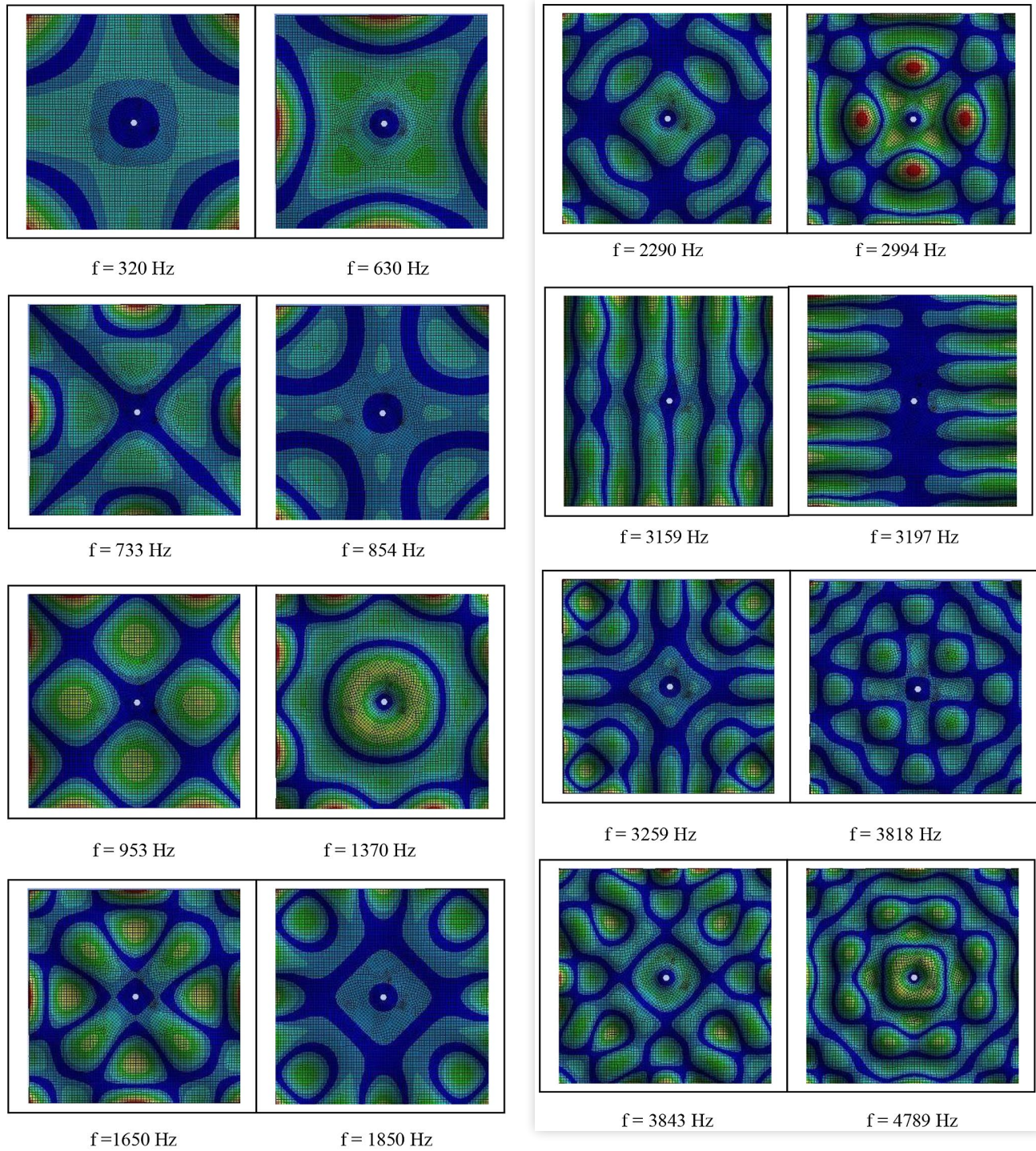


Result and Discussion

Validation of experimental work is done by simulation work. Simulation of chaldni plate is done by ANSYS work bench software and find out percentage error between experimental and simulation work. Solid modal of plates are build in solid works and imported in ANSYS work bench. Messing is done in ANSYS work bench.

Simulation Result for Rectangular Plate

FIGURE 6 Simulation mode shapes for rectangular plate



Simulation Result for Circular Plate

FIGURE 7 Frequency = 560 Hz

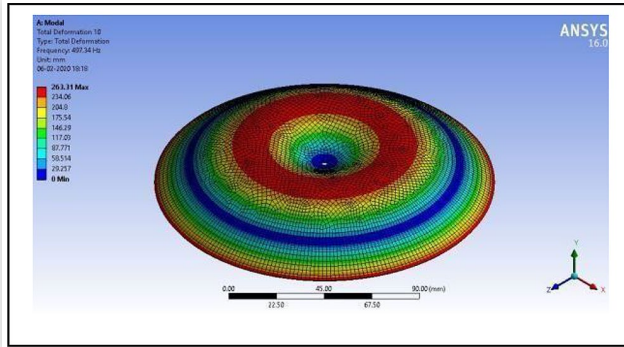


TABLE 2 Percentage error between experimental and simulation work for circular plate

S. N.	Experimental frequency	Simulation frequency	% error
1.	560 Hz	497.34 Hz	11.18
2.	1560 Hz	1445.40 Hz	7.34
3.	2900 Hz	2860.40 Hz	1.36

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FIGURE 8 Frequency = 15600 Hz

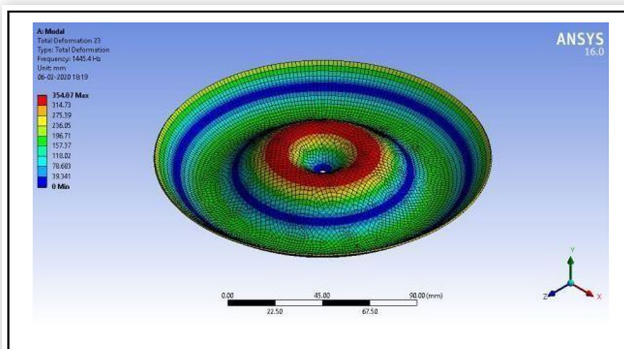


FIGURE 9 Frequency = 2900 Hz

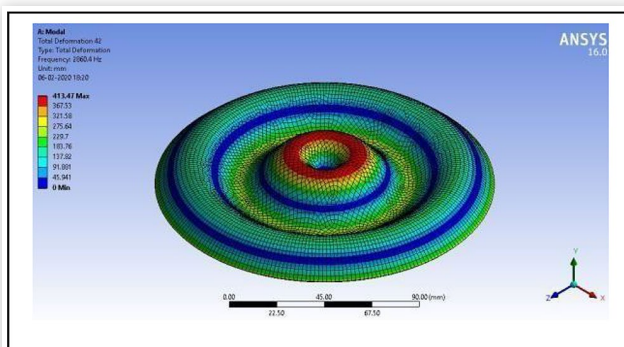


TABLE 1 Percentage error between experimental and simulation work for rectangular plate

S.N	Experimental frequency	Simulation frequency	% error
1.	340	320	5.88
2.	660	630	4.54
3.	760	733	3.55
4.	900	854	4.60
5.	1000	953	4.70
6.	1450	1370	5.51
7.	1770	1650	6.77
8.	1940	1850	4.63
9.	2540	2290	9.84
10.	3250	2994	7.87
11.	3360	3159	5.98
12.	3430	3197	6.79
13.	3480	3259	6.35
14.	3910	3818	2.35
15.	4178	3843	8.01
16.	4987	4789	3.97

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Conclusions

In this paper, Mode shape gives information about the shape of vibration at the particular frequency and acoustic behavior of sound & vibration. Node is the point where the amplitude of vibration is minimum and anti-node is the point where the amplitude of vibration is maximum. For rectangular and circular plates obtained different mode shapes. Mode shape depends on the shape of the plate as the waves go out from the center and rebound back off the sides, and then goes back-reflected different parts, and where the incoming wave is of the same phase but 180 degrees off, there is phase cancellation, and therefore stagnation, where the tea powder accumulates. For rectangular plate obtained 16 mode shapes up to frequency 5000 Hz. More complicated shapes are formed at a higher frequency compared with lower frequency. For circular plate obtained 3 mode shapes up to frequency 3000 Hz. The number of circles is increased with increasing frequency on the circular plate. Various mode shapes are obtained at various frequencies for the rectangular and circular plates. Validation is done by ANSYS workbench software. The percentage error between experimental work and simulation work is less than 15 % for the rectangular and circular plates.

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